KAIZEN MATHS

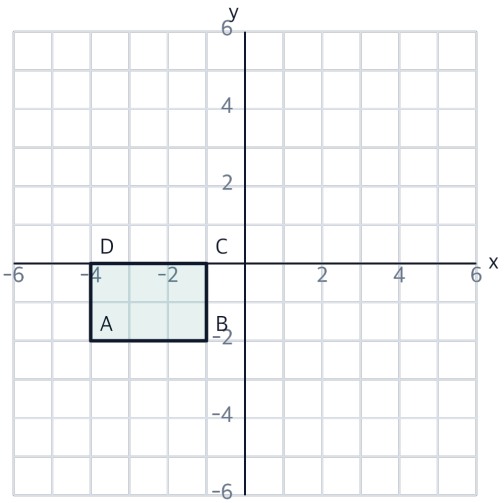
Lesson starter

Transformations

Translate & Reflect

Name _____

SA1



Translate shape ABCD by the vector $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$. Give the coordinates of the image.

GEO-C8E6-L1 · SA

Show your working

KAIZEN MATHS

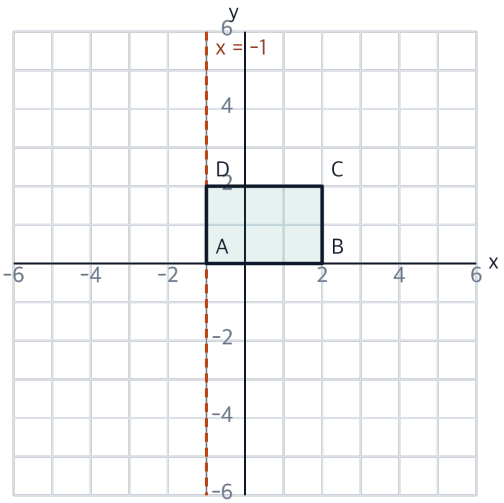
Lesson starter

Transformations

Translate & Reflect

Name _____

SB1



Reflect shape ABCD in the line x = -1. Give the coordinates of the image.

GEO-C8E6-L1 · SB

Show your working

KAIZEN MATHS

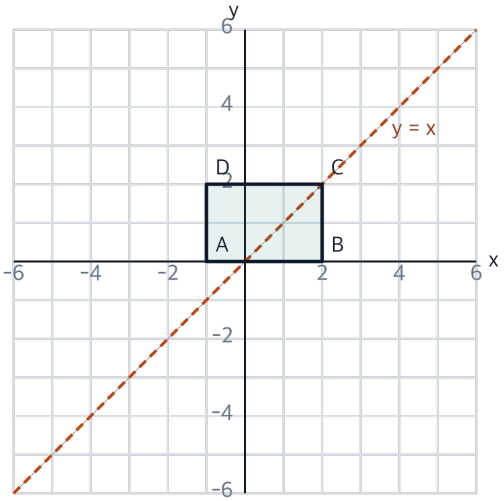
Lesson starter

Transformations

Translate & Reflect

Name _____

SC1



Reflect shape ABCD in the line y = x. Give the coordinates of the image.

GEO-C8E6-L1 · SC

Show your working

KAIZEN MATHS

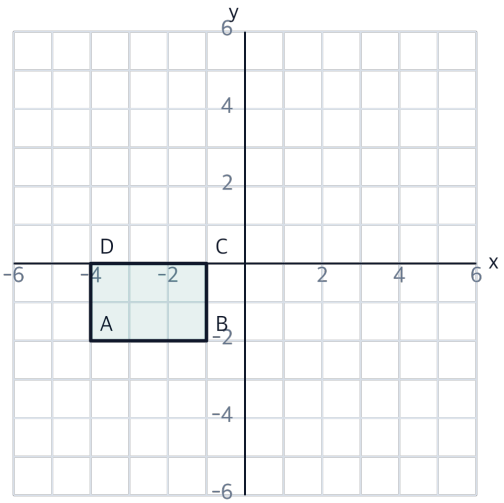
Lesson starter

Transformations

Translate & Reflect

Name _____

SD1



Translate shape ABCD by the vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$. Give the coordinates of the image.

GEO-C8E6-L1 · SD

Show your working

KAIZEN MATHS

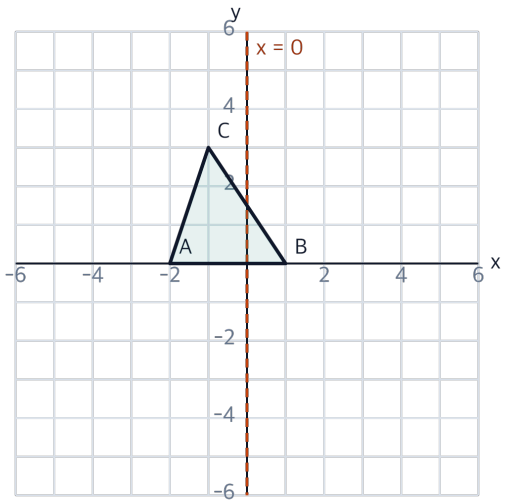
Exit ticket

Transformations

Translate & Reflect

Name _____

EA1



Reflect shape ABC in the line x = 0. Give the coordinates of the image.

GEO-C8E6-L1 · EA

Show your working

KAIZEN MATHS

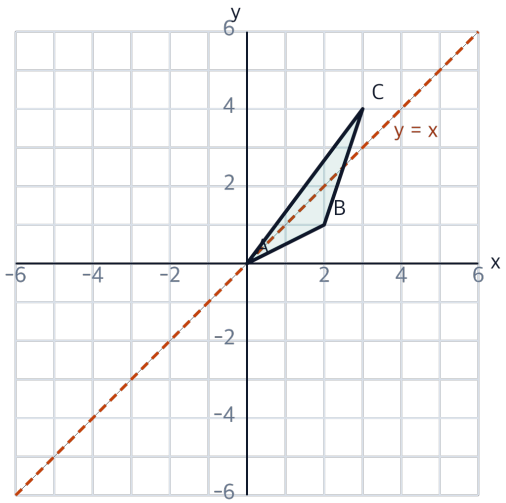
Exit ticket

Transformations

Translate & Reflect

Name _____

EB1



Reflect shape ABC in the line y = x. Give the coordinates of the image.

GEO-C8E6-L1 · EB

Show your working

KAIZEN MATHS

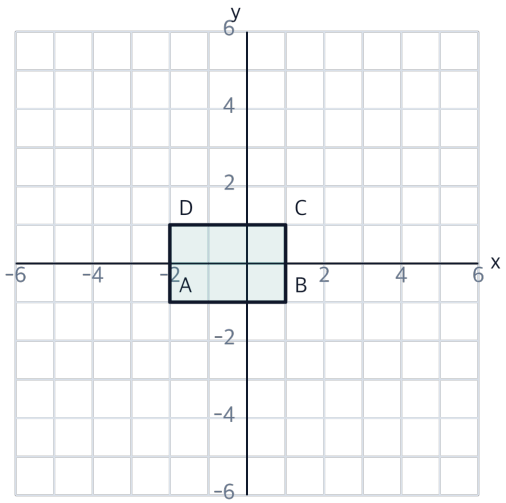
Exit ticket

Transformations

Translate & Reflect

Name _____

EC1



Translate shape ABCD by the vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Give the coordinates of the image.

GEO-C8E6-L1 · EC

Show your working

KAIZEN MATHS

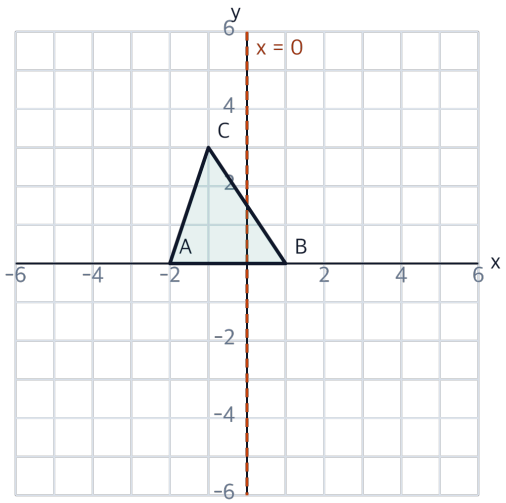
Exit ticket

Transformations

Translate & Reflect

Name _____

EA1



Reflect shape ABC in the line x = 0. Give the coordinates of the image.

GEO-C8E6-L1 · EA

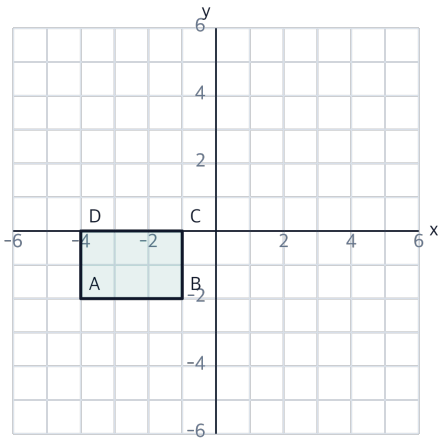
Show your working

Teacher answers & working

Transformations · Translate & Reflect

GEO-C8E6-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Translate shape ABCD by the vector $\begin{pmatrix}-1\\-3\end{pmatrix}$. Give the coordinates of the image.

Answer:

A′(-5, -5), B′(-2, -5), C′(-2, -3), D′(-5, -3)

Working:

A translation adds the same vector to every vertex.

Translation vector = $\begin{pmatrix}-1\\-3\end{pmatrix}$, so the coordinate rule is (x, y) → (x - 1, y - 3).

Vertex	Original point	Rule	Image point
A	(-4, -2)	(x, y) → (x - 1, y - 3)	A′ (-5, -5)
B	(-1, -2)	(x, y) → (x - 1, y - 3)	B′ (-2, -5)
C	(-1, 0)	(x, y) → (x - 1, y - 3)	C′ (-2, -3)
D	(-4, 0)	(x, y) → (x - 1, y - 3)	D′ (-5, -3)

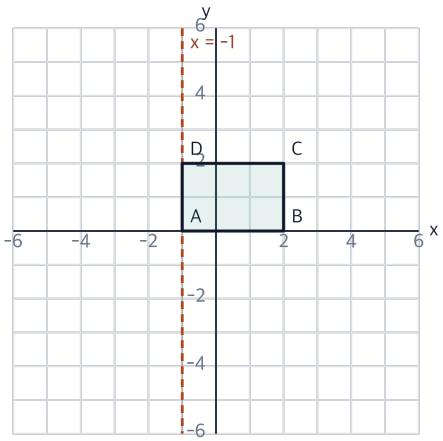
Final image coordinates: A′(-5, -5), B′(-2, -5), C′(-2, -3), D′(-5, -3).

Teacher answers & working

Transformations · Translate & Reflect

GEO-C8E6-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Reflect shape ABCD in the line x = -1. Give the coordinates of the image.

Answer:

A'(-1, 0), B'(-4, 0), C'(-4, 2), D'(-1, 2)

Working:

The mirror line is x = -1.

Each image point is the same perpendicular distance from the mirror line as the original point.

Coordinate rule: (x, y) → (-2 - x, y).

Vertex	Original point	Rule	Image point
A	(-1, 0)	(x, y) → (-2 - x, y)	A' (-1, 0)
B	(2, 0)	(x, y) → (-2 - x, y)	B' (-4, 0)
C	(2, 2)	(x, y) → (-2 - x, y)	C' (-4, 2)
D	(-1, 2)	(x, y) → (-2 - x, y)	D' (-1, 2)

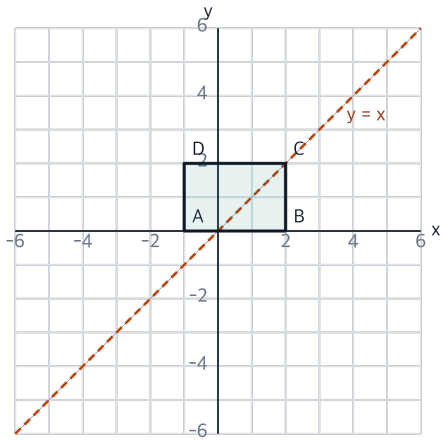
Final image coordinates: A'(-1, 0), B'(-4, 0), C'(-4, 2), D'(-1, 2).

Teacher answers & working

Transformations · Translate & Reflect

GEO-C8E6-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Reflect shape ABCD in the line y = x. Give the coordinates of the image.

Answer:

A′(0, -1), B′(0, 2), C′(2, 2), D′(2, -1)

Working:

The mirror line is y = x.

Reflection in y = x swaps the x-coordinate and y-coordinate.

Vertex	Original point	Rule	Image point
A	(-1, 0)	(x, y) → (y, x)	A′ (0, -1)
B	(2, 0)	(x, y) → (y, x)	B′ (0, 2)
C	(2, 2)	(x, y) → (y, x)	C′ (2, 2)
D	(-1, 2)	(x, y) → (y, x)	D′ (2, -1)

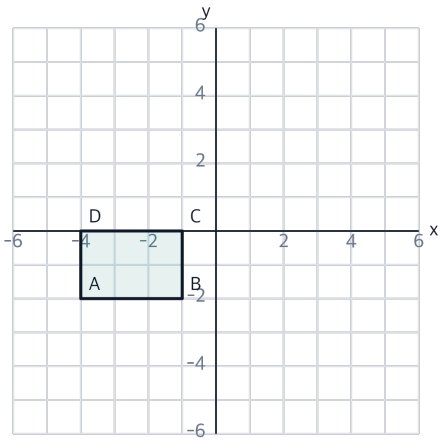
Final image coordinates: A′(0, -1), B′(0, 2), C′(2, 2), D′(2, -1).

Teacher answers & working

Transformations · Translate & Reflect

GEO-C8E6-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Translate shape ABCD by the vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$. Give the coordinates of the image.

Answer:

A′(-2, 0), B′(1, 0), C′(1, 2), D′(-2, 2)

Working:

A translation adds the same vector to every vertex.

Translation vector = $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$, so the coordinate rule is (x, y) → (x + 2, y + 2).

Vertex	Original point	Rule	Image point
A	(-4, -2)	(x, y) → (x + 2, y + 2)	A′ (-2, 0)
B	(-1, -2)	(x, y) → (x + 2, y + 2)	B′ (1, 0)
C	(-1, 0)	(x, y) → (x + 2, y + 2)	C′ (1, 2)
D	(-4, 0)	(x, y) → (x + 2, y + 2)	D′ (-2, 2)

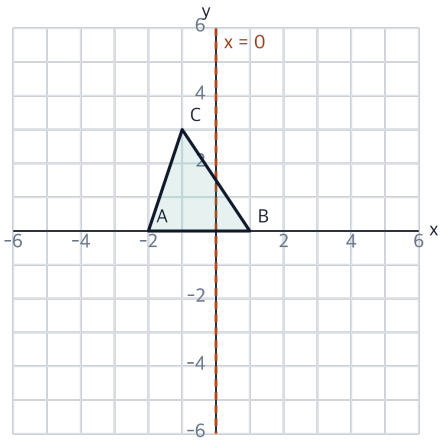
Final image coordinates: A′(-2, 0), B′(1, 0), C′(1, 2), D′(-2, 2).

Teacher answers & working

Transformations · Translate & Reflect

GEO-C8E6-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Reflect shape ABC in the line x = 0. Give the coordinates of the image.

Answer:

A'(2, 0), B'(-1, 0), C'(1, 3)

Working:

The mirror line is x = 0.

Each image point is the same perpendicular distance from the mirror line as the original point.

Coordinate rule: (x, y) → (0 - x, y).

Vertex	Original point	Rule	Image point
A	(-2, 0)	(x, y) → (0 - x, y)	A' (2, 0)
B	(1, 0)	(x, y) → (0 - x, y)	B' (-1, 0)
C	(-1, 3)	(x, y) → (0 - x, y)	C' (1, 3)

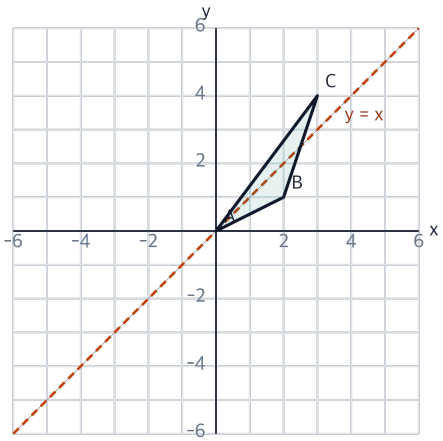
Final image coordinates: A'(2, 0), B'(-1, 0), C'(1, 3).

Teacher answers & working

Transformations · Translate & Reflect

GEO-C8E6-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Reflect shape ABC in the line y = x. Give the coordinates of the image.

Answer:

A′(0, 0), B′(1, 2), C′(4, 3)

Working:

The mirror line is y = x.

Reflection in y = x swaps the x-coordinate and y-coordinate.

Vertex	Original point	Rule	Image point
A	(0, 0)	(x, y) → (y, x)	A′ (0, 0)
B	(2, 1)	(x, y) → (y, x)	B′ (1, 2)
C	(3, 4)	(x, y) → (y, x)	C′ (4, 3)

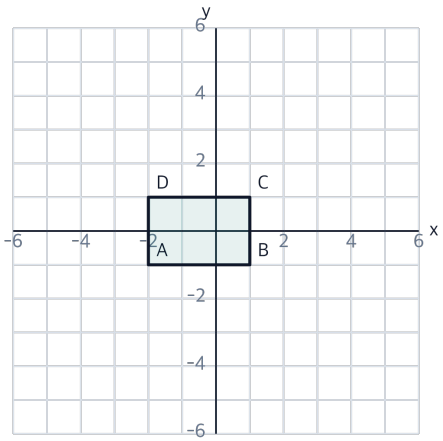
Final image coordinates: A′(0, 0), B′(1, 2), C′(4, 3).

Teacher answers & working

Transformations · Translate & Reflect

GEO-C8E6-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Translate shape ABCD by the vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Give the coordinates of the image.

Answer:

A′(0, 0), B′(3, 0), C′(3, 2), D′(0, 2)

Working:

A translation adds the same vector to every vertex.

Translation vector = $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, so the coordinate rule is (x, y) → (x + 2, y + 1).

Vertex	Original point	Rule	Image point
A	(-2, -1)	(x, y) → (x + 2, y + 1)	A′ (0, 0)
B	(1, -1)	(x, y) → (x + 2, y + 1)	B′ (3, 0)
C	(1, 1)	(x, y) → (x + 2, y + 1)	C′ (3, 2)
D	(-2, 1)	(x, y) → (x + 2, y + 1)	D′ (0, 2)

Final image coordinates: A′(0, 0), B′(3, 0), C′(3, 2), D′(0, 2).